

COLLEGE MANAGEMENT SYSTEM

Submitted By

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1. INTRODUCTION

The College Management System is a backend application developed using Java 17, Spring Boot 3.2.5, MongoDB, Spring Security, JWT Authentication, and Maven. The system provides a centralized platform for managing students, faculty, departments, courses, enrollments, and attendance records.

2. PROJECT OBJECTIVE

To automate and simplify academic administration by digitizing student, faculty, course, enrollment, and attendance management while providing secure access through JWT-based authentication.

3. PROBLEM STATEMENT

Traditional college administration relies on manual records and disconnected systems, leading to inefficiency, duplication, and difficulty in tracking academic information. This project provides a centralized and secure solution.

4. TECHNOLOGY STACK

Language: Java 17

Framework: Spring Boot 3.2.5

Database: MongoDB

Security: Spring Security + JWT (JJWT 0.11.5)

Validation: Jakarta Bean Validation

Build Tool: Maven

Boilerplate Reduction: Lombok

5. SYSTEM ARCHITECTURE

Client → Controller Layer → Service Layer → Repository Layer → MongoDB Database. The Controller handles requests, Service contains business logic, Repository manages database operations, and MongoDB stores application data.

6. DATABASE DESIGN

Collections include Department, Faculty, Student, Course, Enrollment, Attendance, and User. MongoDB document collections are used for storing academic records efficiently.

7. PROJECT STRUCTURE

Main modules include Authentication, Department Management, Faculty Management, Student Management, Course Management, Enrollment System, Attendance Tracking, Security Configuration, Repository Layer, and Service Layer.

8. RELATIONSHIP MAPPING

One Department → Many Faculty
One Department → Many Courses
One Student → Many Enrollments
One Course → Many Enrollments
One Course → Many Attendance Records

9. REST API MODULES

Authentication APIs, Department APIs, Faculty APIs, Student APIs, Course APIs, Enrollment APIs, and Attendance APIs with complete CRUD functionality.

10. TESTING

All REST APIs were tested using Postman. Testing covered Create, Read, Update, Delete operations, authentication, enrollment management, attendance tracking, and validation scenarios.

11. FUTURE ENHANCEMENTS

Email Notifications
Student Portal Dashboard
Faculty Analytics Dashboard
File Upload Support
Advanced Reporting and Analytics
Cloud Deployment Support

12. CONCLUSION

The College Management System successfully demonstrates backend development using Spring Boot and MongoDB. The project implements secure authentication, academic record management, enrollment tracking, attendance management, RESTful APIs, and database integration. It provides a scalable foundation for future academic management solutions.